



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

04-09-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TNAS)

ANSWER KEY

PAPER-1

### PART-1 : PHYSICS

|                 |    |      |       |     |       |     |              |
|-----------------|----|------|-------|-----|-------|-----|--------------|
| SECTION-I (i)   | Q. | 1    | 2     | 3   | 4     |     |              |
|                 | A. | B    | B     | D   | A     |     |              |
| SECTION-I (ii)  | Q. | 5    | 6     | 7   | 8     | 9   | 10           |
|                 | A. | B,C  | A,B,C | A,C | B,C,D | B,D | A,D          |
| SECTION-II (i)  | Q. | 1    | 2     | 3   | 4     | 5   | 6            |
|                 | A. | 0.72 | 7.20  | 7.5 | 8.75  | 5.5 | 1.18 to 1.19 |
| SECTION-II (ii) | Q. | 7    | 8     | 9   |       |     |              |
|                 | A. | 40   | 5     | 20  |       |     |              |

### PART-2 : CHEMISTRY

|                 |    |       |       |       |       |         |       |
|-----------------|----|-------|-------|-------|-------|---------|-------|
| SECTION-I (i)   | Q. | 1     | 2     | 3     | 4     |         |       |
|                 | A. | D     | C     | B     | B     |         |       |
| SECTION-I (ii)  | Q. | 5     | 6     | 7     | 8     | 9       | 10    |
|                 | A. | B,D   | A,C,D | A,B,C | B,C,D | A,B,C,D | A,C,D |
| SECTION-II (i)  | Q. | 1     | 2     | 3     | 4     | 5       | 6     |
|                 | A. | 12.00 | 0.10  | 2.00  | 2.00  | 6.00    | 3.00  |
| SECTION-II (ii) | Q. | 7     | 8     | 9     |       |         |       |
|                 | A. | 44    | 9     | 6     |       |         |       |

### PART-3 : MATHEMATICS

|                 |    |       |       |      |         |      |      |
|-----------------|----|-------|-------|------|---------|------|------|
| SECTION-I (i)   | Q. | 1     | 2     | 3    | 4       |      |      |
|                 | A. | B     | C     | B    | C       |      |      |
| SECTION-I (ii)  | Q. | 5     | 6     | 7    | 8       | 9    | 10   |
|                 | A. | A,D   | A,B,D | B,C  | A,B,C,D | A,B  | C,D  |
| SECTION-II (i)  | Q. | 1     | 2     | 3    | 4       | 5    | 6    |
|                 | A. | -2.00 | 3.00  | 1.00 | -3.00   | 0.00 | 2.00 |
| SECTION-II (ii) | Q. | 7     | 8     | 9    |         |      |      |
|                 | A. | 100   | 6     | 20   |         |      |      |

### HINT – SHEET

#### PART-1 : PHYSICS

##### SECTION-I (i)

##### 1. Ans (B)

Electric field between the plates a

$$f(t) = \frac{v}{\ell} = \frac{kt}{\ell}$$

$$\Rightarrow a = \frac{eE}{m} = \frac{ekt}{m\ell}$$

$$\frac{dv}{dt} = \frac{ekt}{m\ell}$$

$$v = \frac{1}{2} \frac{ekt^2}{m\ell}$$

##### 2. Ans (B)

$$E_x = -\alpha y$$

$$E_y = -\alpha x$$

$$E_z = 2\alpha z$$

$$\text{Hence, } \vec{E} = -\alpha y \hat{i} - \alpha x \hat{j} + 2\alpha z \hat{k}$$

$$\text{Projection of } \vec{E} \text{ along } \vec{A} = \vec{E} \cdot \vec{A}$$

$$= \frac{1}{\sqrt{11}} [-\alpha y - \alpha x + 6\alpha z]$$

$$\text{Putting } x = 2, y = 1, z = -3$$

$$= \frac{1}{\sqrt{11}} [-\alpha - 2\alpha - 18\alpha] = \frac{-21\alpha}{\sqrt{11}}$$

3. Ans (D)

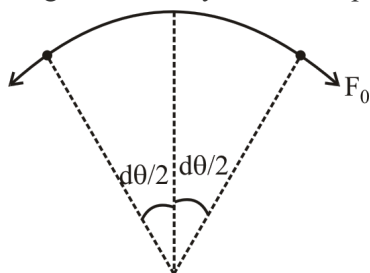
$$\frac{\partial y}{\partial t} = 15 \times 15\pi \sin(\pi x - 15\pi t)$$

$$\cos(\pi x - 15\pi t) = \frac{12}{15} = \frac{4}{5}$$

$$\frac{\partial y}{\partial t} = 225\pi \times \frac{3}{5} = 135\pi \text{ cm/s}$$

4. Ans (A)

If we consider a single wire length  $d\ell$  somewhere in the middle of solenoid then magnetic field by rest of the part  $B = \mu_0 nI$



$$\therefore 2F_0 \sin\left(\frac{d\theta}{2}\right) = Id\ell B$$

$$\Rightarrow 2F_0 \frac{d\theta}{2} = I \cdot \mu_0 nI \cdot R d\theta$$

$$\Rightarrow F_0 = \mu_0 nI^2 R$$

$$\Rightarrow I_{\max} = \sqrt{\frac{F_0}{\mu_0 nR}}$$

## PART-1 : PHYSICS

### SECTION-I (ii)

5. Ans (B,C)

$$\frac{mv}{qB} = r \dots\dots(i)$$

$$\text{For capacitor } E(r) = \frac{V_0}{\ln\left(\frac{b}{a}\right)r}$$

$$\therefore \frac{qv_0}{\ln\left(\frac{b}{a}\right)r} = \frac{mv^2}{r} \Rightarrow \frac{q}{m} = \frac{v^2 \ln\left(\frac{b}{a}\right)}{V_0}$$

$$\text{Also in equation (i) } v = \left(\frac{q}{m}\right) B \cdot r$$

$$\Rightarrow v^2 = \left(\frac{q}{m}\right)^2 B^2 \cdot r^2$$

$$\Rightarrow \left(\frac{q}{m}\right) = \frac{\ln\left(\frac{b}{a}\right)}{V_0} \left(\frac{q}{m}\right)^2 B^2 \cdot r^2$$

$$\Rightarrow \frac{q}{m} = \frac{V_0}{B^2 r^2 \ln\left(\frac{b}{a}\right)}$$

$$\text{And } V = \frac{V_0}{B^2 r^2 \ln\left(\frac{b}{a}\right)} \cdot Br = \frac{V_0}{Br \ln\left(\frac{b}{a}\right)}$$

6. Ans (A,B,C)

$$\text{At resonance } Z = R \Rightarrow I = \frac{V}{R}$$

$$\text{Resonance frequency } f = \frac{1}{2\pi\sqrt{LC}} = 500\text{Hz}$$

$$\text{and } V_L = V_C \Rightarrow V_{LC} = V_L - V_C$$

and voltage across capacitor is  $180^\circ$  out of phase.

7. Ans (A,C)

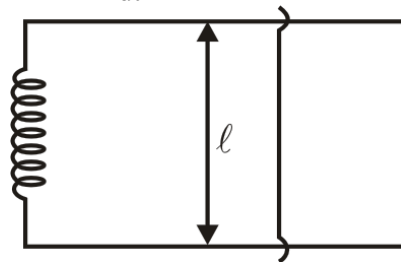
$$B\ell v = \frac{Ldi}{dt} \dots\dots(i)$$

$$F - Bi\ell = \frac{mdv}{dt} \dots\dots(ii)$$

Differentiating equation (ii)

$$-B\ell \frac{di}{dt} = m \frac{d^2v}{dt^2}$$

$$\text{Putting } \frac{di}{dt} = \frac{d\ell v}{L}$$



$$\frac{-B^2 \ell^2}{mL} v = \frac{d^2v}{dt^2}$$

$$\omega^2 = \frac{B^2 \ell^2}{mL}$$

$$\omega = \frac{B\ell}{\sqrt{mL}}$$

8. Ans (B,C,D)

$$v = \frac{12\pi}{\pi/2} = \sqrt{\frac{200}{\mu}}$$

$$\mu = \frac{200}{576}$$

9. Ans (B,D)

$$v(x) = \sqrt{\frac{\left(3m + \frac{m}{\ell}x\right)g}{m}} \times \ell$$

$$\frac{dx}{dt} = \sqrt{(3\ell + x)g} = \int_0^\ell \frac{dx}{\sqrt{(3\ell + x)}} = \sqrt{gt}$$

$$2\sqrt{3\ell + x}\Big|_0^\ell = \sqrt{gt}$$

$$2[2 - \sqrt{3}] \sqrt{\frac{\ell}{g}} = t$$

10. Ans (A,D)

$$T = 2\pi\sqrt{\frac{3}{3\pi^2}} = 2s$$

$$t_1 = \frac{T}{2}$$

$$t_2 = \frac{T}{4}$$

## PART-1 : PHYSICS

### SECTION-II (i)

1. Ans (0.72)

$$Q = \sigma [\pi r^2 + 2\pi rh]$$

$$= 2 \times 10^{-6} \times \pi \left[ \left( \frac{2}{10} \right)^2 + 2 \left( \frac{2}{10} \right) \left( \frac{2}{10} \right) \right]$$

$$= \frac{24\pi}{100} \times 10^{-6}$$

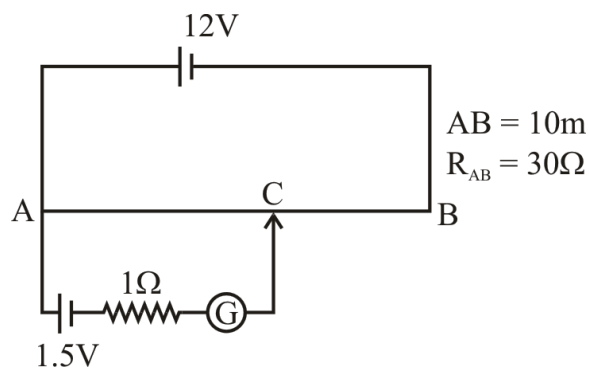
$$\simeq 0.72\mu C$$

2. Ans (7.20)

$$U = \frac{Q^2}{2C} = \frac{(72 \times 10^{-8})^2}{2 \times 36 \times 10^{-12}}$$

$$= 72 \times 10^{-4} \text{ J} = 7.2 \text{ mJ}$$

3. Ans (7.5)



Potential gradient across wire,

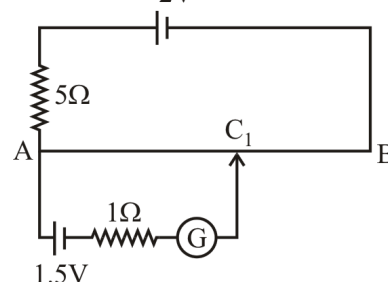
$$AB = \frac{2}{10} = 0.2 \text{ V/m}$$

$$\text{Now, } V_{AC} = 1.5 \text{ V or } (0.2)(AC) = 1.5$$

$$\therefore AC = 7.5 \text{ m}$$

4. Ans (8.75)

$$(a) V_{AB} = \left( \frac{R_{AB}}{R_{AB} + 5} \right) \times 2 = \left( \frac{30}{30 + 5} \right) \times 2 = \frac{12}{7} \text{ V}$$



$\therefore$  Potential gradient across

$$AB = \frac{12}{70} \text{ V/m}$$

$$\text{Now, } V_{AC} = 1.5 \text{ V}$$

$$\therefore \left( \frac{12}{70} \right) (AC_1) = 1.5$$

$$\therefore AC_1 = 8.75 \text{ m}$$

6. Ans (1.18 to 1.19)

(15 & 16)

$$Z_1 = \frac{115}{3} = 38.33\Omega$$

$$\cos \phi_1 = \frac{R_1}{Z_1} \Rightarrow R_1 = Z_1 \cos \phi_1$$

$$\therefore (38.33)(0.6) = 23 \Omega$$

$$X_L = \sqrt{Z_1^2 - R_1^2} = 30.67\Omega$$

$$Z_2 = \frac{115}{5} = 23\Omega$$

$$R_2 = Z_2 \cos \phi_2 = (23)(0.707) = 16.26$$

$$X_C = \sqrt{Z_2^2 - R_2^2} = \sqrt{(23)^2 - (16.26)^2} = 16.26 \Omega$$

When connected in series.

$$R = R_1 + R_2 = 39.26 \Omega$$

$$X_L - X_C = 14.41 \Omega$$

$$\therefore Z = \sqrt{R^2 + (X_L - X_C)^2} = 41.82\Omega$$

$$i = \frac{V}{Z} = \frac{230}{41.82} = 5.5 \text{ A}$$

$$P = i^2 R = (5.5)^2 (39.26) = 1187.6 \text{ W} \approx 1.188 \text{ kW}$$

## PART-1 : PHYSICS

### SECTION-II (ii)

7. Ans (40)

$$B = 10 \log_{10} \left( \frac{I}{10^{-12}} \right)$$

$$I = \frac{4\pi \times 10^{-6}}{4\pi \times 10^2} = 10^{-8}$$

$$B = 10 \log_{10} 10^4 = 40 \text{ dB.}$$

8. Ans (5)

$$\vec{F} = i\vec{l} \times \vec{B}$$

$$\int dF = \int_{x=R}^{\sqrt{5}R} \frac{\mu_0 I}{2\pi x} \times dx \times 2I$$

$$\Rightarrow F_{\text{net}} = \frac{\mu_0 I}{2\pi} \times (2I) \times \ln(\sqrt{5})$$

$$= \frac{\mu_0 I^2}{2\pi} \ln(5)$$

9. Ans (20)

Flux remains constant.

$$B_0 \pi r^2 = B_0 (1+x) \pi r^2 - Li$$

$$\Rightarrow i = \frac{B_0 \pi r^2}{L} x$$

$$\therefore F = i\ell B = \frac{B_0 \pi r^2}{L} x \times 2\pi r \times (2B_0)$$

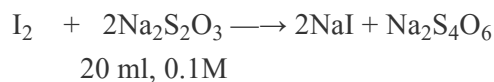
$$\Rightarrow m \frac{v dv}{dx} = F$$

$$\Rightarrow x_{\text{max}} = \left( \frac{v_0^2 m L}{4 B_0^2 \pi^2 r^3} \right)^{1/2} = 0.2 \text{ m} = 20 \text{ cm}$$

## PART-2 : CHEMISTRY

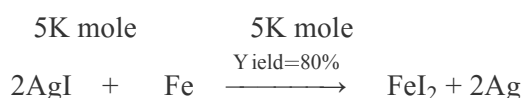
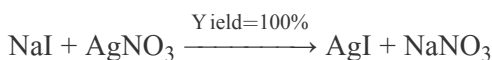
### SECTION-I (ii)

5. Ans (B,D)

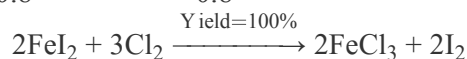


1 m mole 2 m mole

$\therefore I_2 \text{ produced} = 2 \times 10^4 \times 100 \text{ m mole} = 2000 \text{ mole} = 2 \text{ K mole.}$



$$\frac{4}{0.8} = 5K \text{ mole} \quad \frac{2}{0.8} = 2.5K \text{ mole} \quad 2K \text{ mole}$$



2K mole 2K mole

$\therefore AgNO_3 \text{ initially taken} = 5 \text{ K mole}$

$= 5 \times 170 \text{ kg} = 850 \text{ kg.}$

Fe required = 2.5 K mole = 2500 mole.

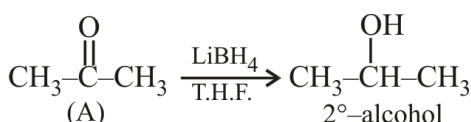
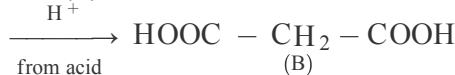
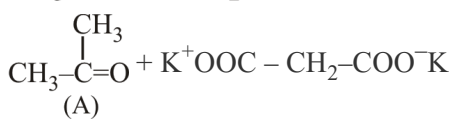
6. Ans (A,C,D)

For expansion and compression both with proper sign conversion

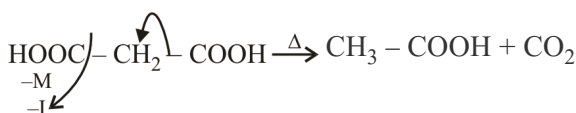
$$W_{\text{irr}} > W_{\text{rev}}$$

Also, for isothermal  $q = -W$ .

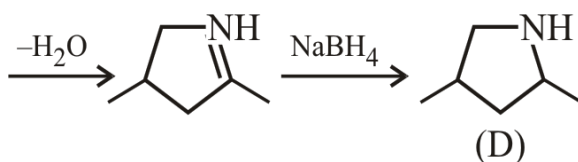
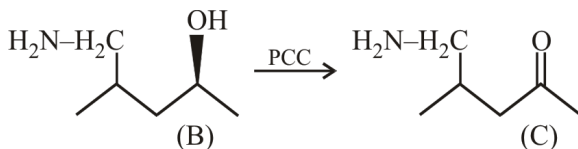
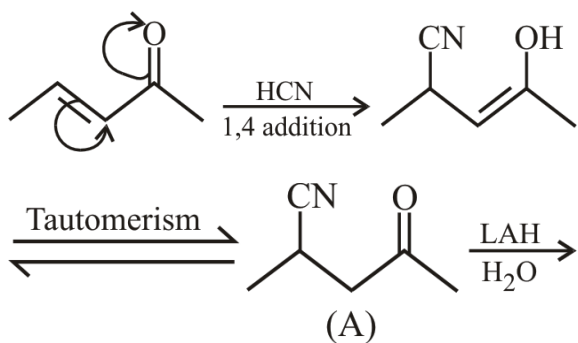
7. Ans (A,B,C)



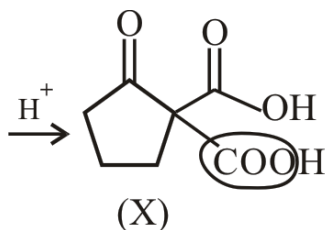
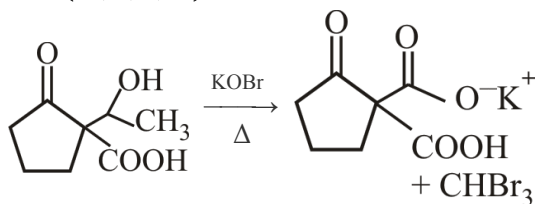
give  
Iodoform  
test



8. Ans (B,C,D)



9. Ans (A,B,C,D)

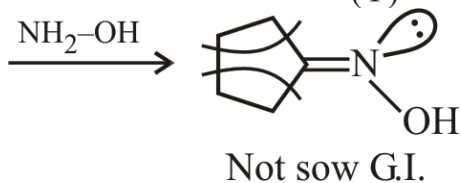
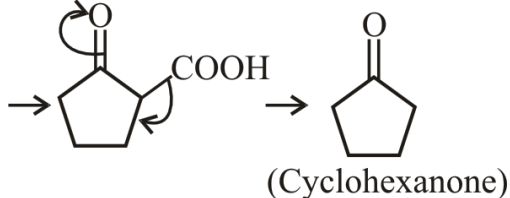


(No C.C.)

2-oxo-cyclopentane-1,1

-Dicarboxylic acid

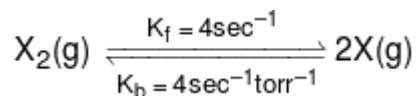
D.W. = 4



## PART-2 : CHEMISTRY

### SECTION-II (i)

2. Ans (0.10)



Initial moles 1 0

moles at eq. 1- $\alpha$  2 $\alpha$  P = 2 torr

$$\frac{r_{\text{mix}}}{r_{\text{He}}} = \frac{1}{1.5} = \sqrt{\frac{4}{M_{\text{mix}}}} \Rightarrow M_{\text{mix}} = 9$$

$$\therefore n \propto \frac{1}{M}$$

$$\frac{1}{1+\alpha} = \frac{9}{M} \Rightarrow 1+\alpha = \frac{M}{9} \dots\dots\dots (1)$$

For elementary reaction,

$$\frac{dP_{X_2}}{dt} = -K_f P_{X_2} + K_b P_X^2$$

$$\text{At equilibrium, } \frac{dP_{X_2}}{dt} = 0$$

$$\therefore K_f P_{X_2} = K_b P_X^2$$

$$\text{or } \frac{K_f}{K_b} = \frac{P_X^2}{P_{X_2}}$$

$$\text{or } \frac{4}{4} = \frac{P_X^2}{P_{X_2}}$$

$$\text{or } 1 = \frac{\left(\frac{2\alpha}{1+\alpha} \times 2\right)^2}{\left(\frac{1-\alpha}{1+\alpha} \times 2\right)} = \frac{8\alpha^2}{1-\alpha^2}$$

$$\text{or } 1 - \alpha^2 = 8\alpha^2 \Rightarrow 1 = 9\alpha^2 \Rightarrow \alpha = \frac{1}{3}$$

$$\therefore \text{from (1), } M = 9 \left(1 + \frac{1}{3}\right) = 9 \times \frac{4}{3} = \boxed{12}$$

Now reaction is irreversible,  $K_f = 4 \text{ sec}^{-1}$ .

$$t_{1/3} = \frac{1}{K} \ln \frac{1}{\left(1 - \frac{1}{3}\right)} = \frac{1}{4} \ln \frac{1}{2/3}$$

$$= \frac{1}{4} \ln \frac{3}{2} = \frac{0.4}{4} = 0.1 \text{ sec.}$$

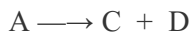
6. Ans (3.00)

(a), (b), (c)

## PART-2 : CHEMISTRY

### SECTION-II (ii)

7. Ans ( 44 )



$$t = 0 \quad 8 \quad 0 \quad 0$$

$$t = 100 \text{ min.} \quad 2 \quad 6 \quad 6$$

$$n_{\text{total}} = 2 + 6 + 6 + 4 + 2 = 20 \text{ mole.}$$

$$P_T = P_A^\circ X_A + P_B^\circ X_B + P_C^\circ X_C + P_D^\circ X_D$$

$$= 90 \times \frac{2}{20} + 40 \times \frac{4}{20} + 50 \times \frac{6}{20} + 40 \times \frac{6}{20}$$

$$= 9 + 8 + 15 + 12 = \boxed{44 \text{ torr}}$$

8. Ans ( 9 )

$$\lambda (\text{in } \text{\AA}) = \sqrt{\frac{150}{V (\text{in volt})}}$$

$$\text{or } 5 = \sqrt{\frac{150}{V}} \text{ or } V = \frac{150}{25} = 6 \text{ volt.}$$

$$\therefore \text{KE of electron} = 6 \text{ eV.}$$

$$\therefore \frac{2\pi r}{\lambda} = n \Rightarrow n = 2$$

$$\therefore \text{electron is in } 2^{\text{nd}} \text{ energy level of He}^+ \text{ ion.}$$

$$E = \frac{-13.62^2}{n^2} = \frac{-13.6 \times 2^2}{2^2} = -13.6 \text{ eV}$$

$$\therefore \text{Energy of photon emitted in this process}$$

$$= 6 - (-13.6) = 19.6 \text{ eV.}$$

$$\therefore E = \phi + (\text{KE})_{\text{max}}$$

$$\text{or } 19.6 = 10.6 + (\text{KE})_{\text{max}}$$

$$\text{or } (\text{KE})_{\text{max}} = 9 \text{ eV.}$$

## PART-3 : MATHEMATICS

### SECTION-I (i)

1. Ans ( B )

$$f(x) = x - 2x^2 + \int_1^x \frac{f(t)}{x} dt$$

$$xf(x) = x^2 - 2x^3 + \int_1^x f(t) dt$$

on differentiation

$$xf'(x) + f(x) = 2x - 6x^2 + f(x)$$

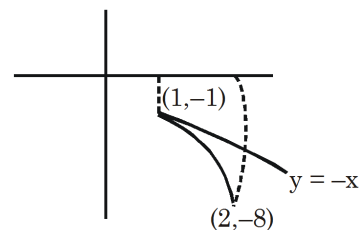
$$f'(x) = 2 - 6x$$

$$\text{on integration } f(x) = 2x - 3x^2 + c$$

$$\therefore f(1) = -1 \Rightarrow c = 0$$

$$f(x) = 2x - 3x^2$$

$$\text{Area} = \int_1^2 [(-x) - (2x - 3x^2)] dx$$



$$= \left[ -\frac{3}{2}x^2 + x^3 \right]_1^2$$

$$= (-6 + 8) - \left( -\frac{3}{2} + 1 \right) = \frac{5}{2}$$

2. Ans ( C )

$$\text{Let } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

$$\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + (a_{12} + a_{21})x_1x_2 + (a_{13} + a_{31})x_1x_3 + (a_{23} + a_{32})x_2x_3 = 0$$

Its true for every  $x_1, x_2, x_3$

$$\text{then } a_{11} = a_{22} = a_{33} = 0, a_{12} + a_{21} = 0,$$

$$a_{13} + a_{31} = 0, a_{23} + a_{32} = 0$$

$\therefore A$  is skew symmetric matrix

$$a_{21} = -2008$$

$$a_{31} = -2010$$

$$a_{32} = 2012$$

3. **Ans (B)**

$$\begin{aligned} \text{Let } t &= \frac{1}{(x+y)^2} \\ \frac{dt}{dx} &= -\frac{2}{(x+y)^3} \cdot \left(1 + \frac{dy}{dx}\right) \\ \text{so } \frac{1}{(x+y)^3} \frac{dy}{dx} + \frac{x}{(x+y)^2} - x^3 &= \frac{-1}{(x+y)^3} \\ \Rightarrow -\frac{1}{2} \frac{dt}{dx} &= -tx + x^3 \\ \frac{dt}{dx} &= 2tx - 2x^3 \\ \text{I.F.} &= e^{\int -2x dx} = e^{-x^2} \\ \Rightarrow \frac{1}{(x+y)^2} &= x^2 + 1 + ce^{x^2} \end{aligned}$$

4. **Ans (C)**

$$\begin{aligned} \text{We have } f(x) &= \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t f(t)) dt \\ &= \sin x + \pi \sin x + \int_{-\pi/2}^{\pi/2} t f(t) dt \\ \therefore f(x) &= (\pi + 1) \sin x + A \quad \dots (1) \\ \text{Now } A &= \int_{-\pi/2}^{\pi/2} t ((\pi + 1) \sin t + A) dt \\ &= 2(\pi + 1) \int_0^{\pi/2} t \sin t dt \quad \begin{matrix} \text{(I)} \\ \text{(II)} \\ \text{(By part)} \end{matrix} \\ \Rightarrow A &= 2(\pi + 1) \\ \text{Hence } f(x) &= (\pi + 1) \sin x + 2(\pi + 1) \\ \therefore f_{\max} &= 3(\pi + 1) = M \\ \text{and } f_{\min} &= (\pi + 1) = m \\ \Rightarrow \frac{M}{m} &= 3 \end{aligned}$$

## PART-3 : MATHEMATICS

### SECTION-I (ii)

5. **Ans (A,D)**

$$\begin{aligned} AB = A &\Rightarrow ABA = A^2 \Rightarrow A^2 = A \\ \text{Similarly } B^2 &= B \\ A \text{ and } B &\text{ are idempotent} \\ \text{Also } (A - B)^2 &= (A - B)(A - B) \\ &= A^2 - AB - BA + B^2 = A - A - B + B = 0 \\ (A - B) &\text{ is nilpotent.} \\ \therefore \text{Correct options are (A) and (D).} \end{aligned}$$

6. **Ans (A,B,D)**

$$\begin{aligned} \int \frac{dx}{x^2 + ax + 1} &= \int \frac{dx}{\left(x + \frac{a}{2}\right)^2 + \left(1 - \frac{a^2}{4}\right)} \\ \text{If } \frac{a^2}{4} < 1 &\text{ then } 1 - \frac{a^2}{4} = k^2 \\ \Rightarrow \int \frac{dx}{\left(x + \frac{a}{2}\right)^2 + k^2} &= \frac{1}{k} \tan^{-1} \left( \frac{x + \frac{a}{2}}{k} \right) + c \\ \text{If } \frac{a^2}{4} > 1 &\text{ then } 1 - \frac{a^2}{4} = -k^2 \\ \Rightarrow \int \frac{dx}{\left(x + \frac{a}{2}\right)^2 - k^2} &= \frac{1}{2k} \ln \left| \frac{x + \frac{a}{2} - k}{x + \frac{a}{2} + k} \right| + c \end{aligned}$$

7. **Ans (B,C)**

$$\begin{aligned} 3ABA^{-1} + A &= 2A^{-1}BA \\ \Rightarrow 3ABA^{-1} + A + 2A &= 2A^{-1}BA + 2A \\ \Rightarrow 3A(BA^{-1} + I) &= 2(A^{-1}B + I)A \\ \Rightarrow 3A(B + IA)A^{-1} &= 2A^{-1}(B + IA)A \\ \Rightarrow 3A(B + A)A^{-1} &= 2A^{-1}(B + A)A \\ \text{Let } B + A &= x \\ \Rightarrow 3Ax A^{-1} &= 2A^{-1}xA \\ \Rightarrow 3^n |A| |x| |A^{-1}| &= 2^n |A^{-1}| |x| |A| \\ \Rightarrow 3^n |x| &= 2^n |x| \quad (\text{as } |A| \neq 0) \\ \Rightarrow |x| &= 0 \text{ or } |A + B| = 0 \\ \text{Let } M &= ABA^{-1} - A^{-1}BA \\ \therefore AM &= A^2BA^{-1} - BA \\ \Rightarrow BA &= A^2BA^{-1} - AM \\ \text{Now } 3ABA^{-1} + A &= 2A^{-1}BA \\ &= 2A^{-1}(A^2BA^{-1} - AM) \\ &= 2ABA^{-1} - 2M \\ \Rightarrow ABA^{-1} + A &= -2M \\ \Rightarrow A(BA^{-1} + I) &= -2M \\ A(A + B)A^{-1} &= -2M \\ \text{Taking determinants both sides} \\ |-2M| &= |A||A + B||A^{-1}| = 0 \\ \Rightarrow |ABA^{-1} - AB^{-1}A| &= 0 \end{aligned}$$

8. Ans (A,B,C,D)

$$f(1-x) = f(1+x)$$

$$\Rightarrow f'(1-x) + f'(1+x) = 0$$

$$\text{put } x = 0 \Rightarrow f'(1) = 0$$

$$\text{Given } f'\left(\frac{1}{4}\right) = 0 \Rightarrow f'\left(\frac{7}{4}\right) = 0$$

$$f'\left(\frac{1}{2}\right) = 0 \Rightarrow f'\left(\frac{3}{2}\right) = 0$$

Applying Rolle's theorem in

$$\left(\frac{1}{4}, \frac{1}{2}\right), \left(\frac{1}{2}, 1\right), \left(1, \frac{3}{2}\right), \left(\frac{3}{2}, \frac{7}{4}\right)$$

$$f''(x) = 0 \text{ has at least 4 roots in } (0, 2)$$

$$\Rightarrow (A)$$

$$I = \int_{-2}^2 (x^5 + x^3) f(1+x) dx$$

$$I = \int_{-2}^2 ((-x)^5 + (-x)^3) f(1-x) dx$$

$$\Rightarrow I + I = \int_{-2}^2 f(1+x) \times 0 dx = 0$$

$$\Rightarrow I = 0 < \int_0^2 \frac{1}{1+2^{f(x)}} dx \quad (B)$$

$$\text{Suppose } F(x) = \frac{f'(x)}{x}$$

$$F(1) = F\left(\frac{3}{2}\right) = 0$$

Applying Rolle's theorem

$$F'(c) = 0 \text{ for at least one } c \text{ in } \left(1, \frac{3}{2}\right)$$

$$\Rightarrow c \cdot f''(c) - f''(c) = 0$$

$$\Rightarrow c \cdot f''(c) = f'(c)$$

$$\text{for at least one } c \text{ in } \left(1, \frac{3}{2}\right)$$

$$\text{Suppose } G(x) = x \cdot f(x)$$

$$G\left(\frac{3}{2}\right) = G\left(\frac{7}{4}\right) = 0$$

$$G'(x) = 1 \cdot f'(x) + x f''(x)$$

$$\text{Applying Rolle's theorem in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

$$\Rightarrow f'(c) + c f''(c) = 0 \text{ for at least one } c \text{ in } \left(\frac{3}{2}, \frac{7}{4}\right)$$

9. Ans (A,B)

$$\frac{d(f(x))^2 + (f'(x))^2}{dx} = 2f'(x)(f(x) + f''(x))$$

$$= -2xg(x)(f(x))^2 < 0 \forall x > 0$$

$$\therefore (f(x))^2 < 25 - (f'(x))^2 < 25$$

$$\therefore |f(x)| < 5.$$

10. Ans (C,D)

$$f(x) - 2\lambda \frac{\sin^2 x}{\cos^5 x} = \frac{\sin^2 x}{\cos^5 x}$$

$$\text{Hence, } \lambda = (2\lambda + 1) \int_0^{\frac{\pi}{4}} \frac{\sin^2 t}{\cos^4 t} dt$$

$$\Rightarrow \lambda = (2\lambda + 1) \int_0^{\frac{\pi}{4}} \tan^2 t \sec^2 t dt \Rightarrow \lambda = 1$$

$$\therefore f(x) = \frac{3\sin^2 x}{\cos^5 x}$$

$$f(0) = 0, \lim_{x \rightarrow \frac{\pi}{2}} f(x) = \infty, \lim_{x \rightarrow \frac{\pi}{2}} f(x) = -\infty,$$

$$f(\pi) = 0$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (-2.00)

$$A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

$$|A| = adf = 1$$

$$\text{Given : } A^3 + \alpha A^2 + \beta A - I = 0$$

$$\text{So } |A| = 1$$

$$C-1 : a = d = f = 1$$

$$A^3 - 3A^2 + 3A - I = 0$$

$$\Rightarrow \alpha = -3, \beta = 3$$

$$C-2 : \text{Any two of } a, d, f = -1 \text{ and other is } 1.$$

$$A^3 + A^2 - A - I = 0 \Rightarrow a = 1, b = -1$$

$$\text{Sum of all the possible value of } \alpha \text{ is } -2$$

$$\text{also when } \alpha = -3 \Rightarrow \beta = 3$$

2. Ans ( 3.00 )

$$A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

$$|A| = adf = 1$$

$$\text{Given : } A^3 + \alpha A^2 + \beta A - I = 0$$

$$\text{So } |A| = 1$$

$$C-1 : a = d = f = 1$$

$$A^3 - 3A^2 + 3A - I = 0$$

$$\Rightarrow \alpha = -3, \beta = 3$$

$$C-2 : \text{Any two of } a, d, f = -1 \text{ and other is } 1.$$

$$A^3 + A^2 - A - I = 0 \Rightarrow a = 1, b = -1$$

$$\text{Sum of all the possible value of } \alpha \text{ is } -2$$

$$\text{also when } \alpha = -3 \Rightarrow \beta = 3$$

3. Ans ( 1.00 )

$$f^2(x) - f^2(y) = 3f(x) - 3f(y) + \sqrt{x} - \sqrt{y}$$

$$2f(x)f'(x) = 3f'(x) + \frac{1}{2\sqrt{x}}$$

$$(2f(x) - 3)f'(x) = \frac{1}{2\sqrt{x}}$$

Integrate w.r.t. 'x'

$$f(x) = \frac{1}{2} \left( 3 + \sqrt{4\sqrt{x} + 1} \right)$$

$$\lim_{x \rightarrow 0^+} \frac{2f(x) - 3 - e^{\sqrt{x}}}{\sqrt{x}} = \lim_{x \rightarrow 0^+} \frac{4\sqrt{x} + 1 - e^{2\sqrt{x}}}{\sqrt{x} (\sqrt{4\sqrt{x} + 1} + 1 + e^{\sqrt{x}})} = 1$$

4. Ans ( -3.00 )

$$g'(3) = \frac{1}{f'(4)} = 12$$

$$\frac{d}{dx} \left( \frac{4}{g(x)} \right) \bigg|_{x=3} = -3$$

5. Ans ( 0.00 )

$$f(x) = \sin x + \cos x + |\sin x - \cos x|$$

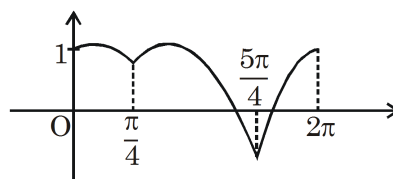
$$= 2\max(\sin x, \cos x)$$

$$\lim_{h \rightarrow 0} \frac{(f(\pi + h))^2 - (f(\pi - h))^2}{h}$$

$$\lim_{h \rightarrow 0} \left( \frac{f(\pi + h) - f(\pi - h)}{2h} \right) (2)(f(\pi + h) + f(\pi - h))$$

$$= f'(\pi) \cdot 2 \cdot (2f(\pi)) = 0$$

6. Ans ( 2.00 )



## PART-3 : MATHEMATICS

### SECTION-II (ii)

7. Ans ( 100 )

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^n = \begin{cases} I & n \text{ is even} \\ A & n \text{ is odd} \end{cases}$$

$$\sum_{r=1}^{100} A^r = (A + A^3 + A^5 + \dots + A^{99}) + (A^2 + A^4 + \dots + A^{100})$$

8. Ans ( 6 )

$$OP = OM$$

$$y - x \frac{dy}{dx} = \sqrt{x^2 + y^2}$$

$$\frac{dy}{dx} = \frac{y - \sqrt{x^2 + y^2}}{x}$$

$$\frac{dy}{dx} = \frac{y}{x} - \sqrt{1 + \left(\frac{y}{x}\right)^2}$$

$$\text{Putting } \frac{y}{x} = v \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore v + x \frac{dv}{dx} = v - \sqrt{1 + v^2}$$

$$\Rightarrow \log(v + \sqrt{1 + v^2}) = \log \frac{c}{x}$$

$$\Rightarrow (v + \sqrt{1 + v^2}) = \frac{c}{x}$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = c$$

Curve is parabola so  $e = 1$

9. Ans ( 20 )

$$f'(x) = f(x) \cdot n(x+1)^{n-1}$$

$$f''(x) = f(x)n(n-1)(x+1)^{n-2} + f'(x)n(x+1)^{n-1}$$

$$f''(1) = n(n-1)f(1)2^{n-2} + f(1)n2^{n-1} \cdot n2^{n-1}$$

$$= 67 \cdot 2^n \cdot e^{2^n}$$

$$n \cdot 2^{n-2} (n-1 + n \cdot 2^n) = 67 \cdot 2^n$$

$$\frac{n}{4} (n-1 + n \cdot 2^n) = 67$$

$$\Rightarrow n = 4, k = 2^n = 16$$



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

04-09-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TNAS)

ANSWER KEY

PAPER-2

### PART-1 : PHYSICS

| SECTION-I (i)   | Q. | 1    | 2     | 3            | 4            | 5    | 6     |
|-----------------|----|------|-------|--------------|--------------|------|-------|
|                 | A. | B,C  | A,C,D | A,C          | B,C          | A,D  | B,C,D |
| SECTION-I (ii)  | Q. | 7    | 8     | 9            | 10           |      |       |
|                 | A. | B    | B     | D            | B            |      |       |
| SECTION-II (i)  | Q. | 1    | 2     | 3            | 4            | 5    | 6     |
|                 | A. | 1.00 | 2.80  | 0.78 to 0.79 | 0.44 to 0.45 | 1.33 | 5.33  |
| SECTION-II (ii) | Q. | 7    | 8     | 9            |              |      |       |
|                 | A. | 0    | 6     | 4            |              |      |       |

### PART-2 : CHEMISTRY

| SECTION-I (i)   | Q. | 1     | 2     | 3                  | 4      | 5     | 6    |
|-----------------|----|-------|-------|--------------------|--------|-------|------|
|                 | A. | B,C,D | A,C,D | A,B,C              | A,B,D  | B,C,D | C,D  |
| SECTION-I (ii)  | Q. | 7     | 8     | 9                  | 10     |       |      |
|                 | A. | D     | B     | B                  | B      |       |      |
| SECTION-II (i)  | Q. | 1     | 2     | 3                  | 4      | 5     | 6    |
|                 | A. | 44.50 | 8.33  | 2450.53 or 2450.54 | 159.92 | 3.00  | 5.00 |
| SECTION-II (ii) | Q. | 7     | 8     | 9                  |        |       |      |
|                 | A. | 3     | 3     | 2                  |        |       |      |

### PART-3 : MATHEMATICS

| SECTION-I (i)   | Q. | 1     | 2     | 3    | 4      | 5     | 6       |
|-----------------|----|-------|-------|------|--------|-------|---------|
|                 | A. | A,B,C | A,B   | A,D  | B,C,D  | A,B,C | A,C     |
| SECTION-I (ii)  | Q. | 7     | 8     | 9    | 10     |       |         |
|                 | A. | B     | A     | C    | B      |       |         |
| SECTION-II (i)  | Q. | 1     | 2     | 3    | 4      | 5     | 6       |
|                 | A. | 3.00  | 53.00 | 0.00 | 255.00 | 51.00 | 4037.00 |
| SECTION-II (ii) | Q. | 7     | 8     | 9    |        |       |         |
|                 | A. | 200   | 6     | 4    |        |       |         |

### HINT – SHEET

#### PART-1 : PHYSICS

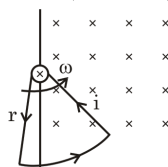
##### SECTION-I (i)

1. Ans (B,C)

$$A = \text{const.}$$

$$v = A\omega.$$

2. Ans (A,C,D)



While entering torque on loop

$$\tau = B i r \cdot \frac{r}{2} (-\hat{k})$$

$$\tau = \frac{B r^2}{2} \left( \frac{B \omega r^2}{2R} \right)$$

$$I \omega \frac{d\omega}{d\theta} = - \frac{B^2 \omega r^4}{4R} \Rightarrow d\omega = - \frac{B^2 r^4}{4IR} \cdot d\theta$$

$$\text{for } \theta = \frac{\pi}{4} \Rightarrow \omega = 1 \text{ rad/s} \Rightarrow \text{time} = \frac{\pi/2}{1} = \frac{\pi}{2} \text{ s}$$

4. Ans (B,C)

$$Z_1 = R_1 + iX_L \cos \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = 45^\circ \Rightarrow R_1 = X_L$$

$$Z_2 = R_2 - iX_C \cos \phi = \frac{1}{\sqrt{2}} \Rightarrow \phi = 45^\circ \Rightarrow R_2 = X_C$$

$$I_{1\max} = \frac{I_{2\max}}{9} \Rightarrow \frac{V}{R_1} = \frac{V}{9R_2} \Rightarrow R_1 = 9R_2$$

$$\text{Series combination} \Rightarrow Z_1 + Z_2 = R_1 + R_2 + i(X_L - X_C) = 10R_2 + i8R_2$$

$$\cos \phi = 0.78$$

5. Ans (A,D)

Flux linked to ring = 0

$$(B_{\text{ext}} + B_{\text{induced}})\pi r^2 = 0$$

$$\Rightarrow 3 \left( \frac{\mu_0}{4\pi} \frac{qv \sin 30^\circ}{(a/\sqrt{3})^2} \right) - \frac{\mu_0 i}{2r} = 0$$

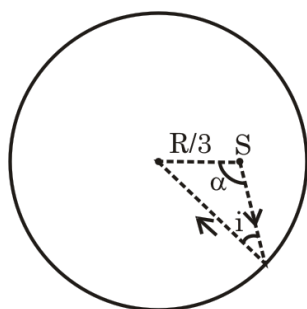
$$i = \frac{9}{4} \frac{qvr}{\pi a^2}$$

$$\text{Magnetic moment} = M = i\pi r^2$$

$$\text{Force on any charge} = qvB_{\text{ring}} \sin 90^\circ$$

$$= qv \frac{\mu_0 M}{4\pi (a/\sqrt{3})^3} \Rightarrow F \propto \frac{1}{a^5}$$

6. Ans (B,C,D)



$$-\frac{7}{4R} - \frac{1.5}{\mu} = \frac{1-1.5}{-R} \Rightarrow u = -\frac{2R}{3}$$

$$\text{Distance from center} = R - \frac{2R}{3} = \frac{R}{3}$$

for any incident ray

$$\frac{\sin \alpha}{R} = \frac{\sin i}{R/3}$$

$$\Rightarrow \sin i = \frac{\sin \alpha}{3}$$

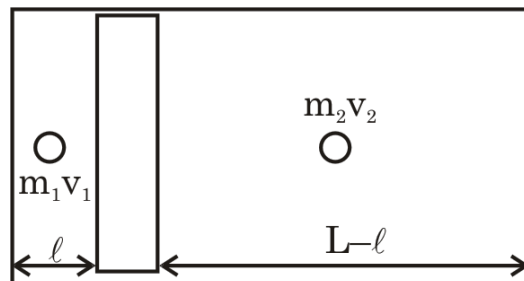
$$\text{for } i_{\max} \Rightarrow \alpha = 90^\circ \Rightarrow i_{\max} = \sin^{-1} \left( \frac{1}{3} \right)$$

$$i_{\max} < \theta_c$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (B)



Force on piston by any ball

$$= \frac{2mv}{2\ell/v} = \frac{mv^2}{\ell}$$

In equilibrium

$$\frac{m_1 v_1^2}{\ell} = \frac{m_2 v_2^2}{L - \ell}$$

$$\text{when } m_1 = m_2 \text{ and } v_1 = v_2 \Rightarrow \ell = \frac{L}{2}$$

$$\text{and when } m_1 = \frac{m_2}{2} \text{ and } v_1 = \frac{v_2}{2}$$

$$\Rightarrow \frac{1}{8\ell} = \frac{1}{L - \ell} \Rightarrow \ell = \frac{L}{9}$$

when piston is displaced by small distance x

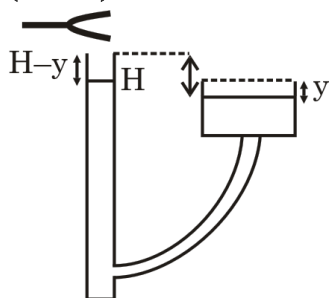
net force is

$$= \frac{mv^2}{\frac{L}{2} + x} - \frac{mv^2}{\frac{L}{2} - x} \approx -\frac{2mv^2 x}{\frac{L^2}{4}}$$

$$F \Rightarrow -\frac{8mv^2}{L^2} x$$

$$\text{Time period} = 2\pi \sqrt{\frac{ML^2}{8mv^2}} = 20 \text{ sec}$$

10. Ans ( B )  
( 9 & 10 )



$$6Ay = A(H-y)$$

$$\Rightarrow 7y = H$$

$$y = \frac{H}{7}$$

$$\Delta H = H - \frac{H}{7} = \frac{6H}{7}$$

$$\frac{\lambda}{4} = \frac{6}{7} \times 21 + e$$

$$\frac{3\lambda}{4} = \frac{6}{7} \times 70 + e$$

$$\frac{\lambda}{2} = 42 \Rightarrow \lambda = 84 \text{ cm}$$

$$v = f\lambda = 336 \text{ m/s}$$

$$e = 3 \text{ cm}$$

$$e = 3 \text{ cm} = 0.6 \text{ r}$$

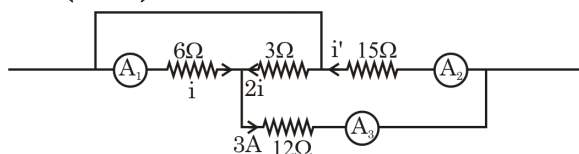
$$r = 5 \text{ cm}$$

$$A = 6 \times \pi \times 5^2 = 150 \pi \text{ cm}^2$$

## PART-1 : PHYSICS

### SECTION-II (i)

2. Ans ( 2.80 )



$$i + 2i = 3$$

$$i = 1 \text{ A}$$

$$\text{KVL} \Rightarrow 3 \times 2i + 3 \times 12 + i' \times 15 = 0$$

$$\Rightarrow i' = -2.8 \text{ A}$$

4. Ans ( 0.44 to 0.45 )

$$\text{Number of prisms} = \frac{x}{t}$$

$$\text{Total deviation} = \frac{x}{t} \cdot (\mu - 1) A = \frac{\pi}{4} x = \theta$$

$$\text{Slope of ray} \Rightarrow \frac{dy}{dx} = -\tan\left(\frac{\pi}{4}x\right)$$

$$\Rightarrow y_0 - y = \frac{4}{\pi} \ln\left(\sec\left(\frac{\pi}{4}x\right)\right)$$

6. Ans ( 5.33 )

System acquires steady state immediately because there is no resistance.

In steady state

$$Bv\ell = \frac{q}{C}$$

During charge redistribution

$$F = -Bil = ma$$

$$\Rightarrow -B \frac{dq}{dt} \ell = m \frac{dV}{dt}$$

$$\Rightarrow V = V_0 - \frac{Bq\ell}{m}$$

$$\Rightarrow V = V_0 - \frac{B\ell}{m} (Bv\ell C)$$

$$V = \frac{V_0}{1 + \frac{B^2 \ell^2 C}{m}} = \frac{4}{3}$$

$$\text{Distance} = \frac{4}{3} \times 1 = \frac{4}{3} \text{ m}$$

$$\text{Energy lost} = -\Delta E = -(\Delta K + \Delta U)$$

$$= \left( \frac{1}{2} mv_0^2 - \frac{1}{2} mv^2 \right) - \frac{q^2}{2C}$$

$$= \frac{1}{2} mv_0^2 \left( 1 - \frac{V}{V_0} \right) = \frac{16}{3} \text{ J}$$

## PART-1 : PHYSICS

### SECTION-II (ii)

7. Ans ( 0 )

Switch closing will not disturb the circuit.

8. Ans ( 6 )

$$a_P = V_\omega \left( \frac{\partial V_P}{\partial x} \right) = f\lambda (-\tan 60) = -6 \text{ m/s}^2$$

9. Ans ( 4 )

$$E = \frac{n\pi^2}{2} \mu v A^2 f$$

$$= 4 \times \frac{\pi^2}{2} \mu \sqrt{\frac{T}{\mu}} \times A^2 \times \frac{4}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$= \frac{4\pi^2 T A^2}{\ell} = \frac{4\pi^2}{3} \times 30 \times 10^{-8} = 4\mu\text{J}$$

## PART-2 : CHEMISTRY

### SECTION-I (i)

1. Ans ( B,C,D )

If  $T > T_B$  then  $z > 1$

If  $T < T_B$  then  $z < 1$

2. Ans (A,C,D)

$$r = K [H^+] [\text{ester}]$$

$$0.1 = K \times 0.1 \times 0.2 \dots\dots\dots (1)$$

$$\Rightarrow K = \frac{0.1}{0.1 \times 0.2} = 5$$

$$10^{-3} = K \times [H^+]_{HB} \times 0.4 \dots\dots\dots (2)$$

$$\frac{(1)}{(2)} \Rightarrow 100 = \frac{0.1}{[H^+]_{HB}} \times \frac{1}{2}$$

$$\text{or } [H^+]_{HB} = \frac{0.1}{100 \times 2} = \frac{10^{-3}}{2}$$



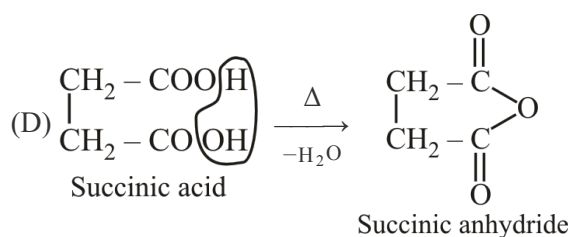
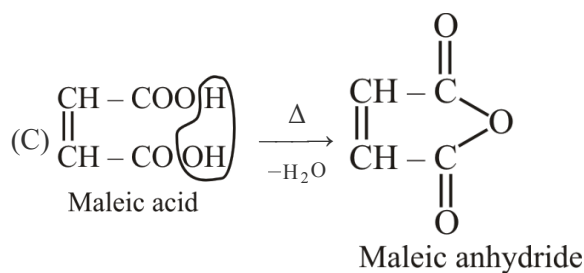
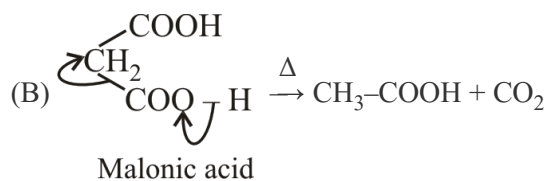
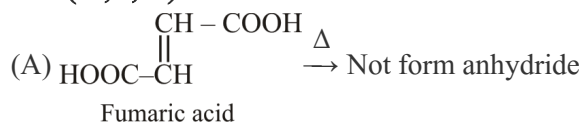
$$1 \quad \frac{10^{-3}}{2} \quad \frac{10^{-3}}{2}$$

$$K_a = \frac{10^{-6}}{4} = 0.25 \times 10^{-6} = 2.5 \times 10^{-7}$$

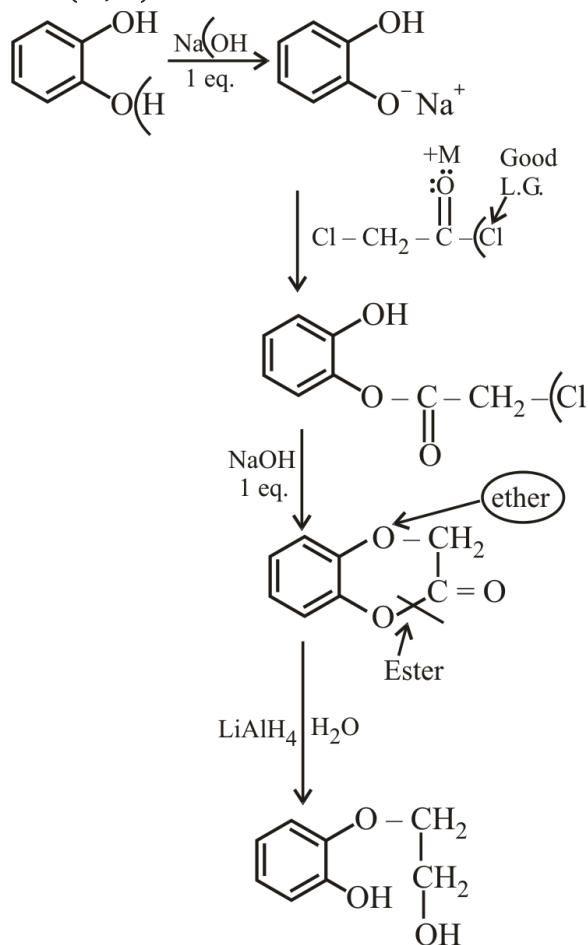
$$K_{HB} = K \times [H^+]_{HB} = \frac{5 \times 10^{-3}}{2} = 2.5 \times 10^{-3} \text{ sec}^{-1}$$

$$t_{1/2} = \frac{\ln 2}{K_{HB}} = \frac{0.7}{2.5 \times 10^{-3}} = \frac{100 \times 7}{2.5} = 280 \text{ sec.}$$

5. Ans (B,C,D)



6. Ans (C,D)



PART-2 : CHEMISTRY

SECTION-II (i)

2. Ans (8.33)

$$\text{HCl} = \frac{567.5}{22700} = 0.025 \text{ mole} = 25 \text{ m mole.}$$

$$\text{BOH} = \frac{0.3075}{M} \times 1000 \text{ m mole} = \frac{307.5}{M} \text{ m mole}$$

$$\text{pH} = 1.4 \Rightarrow [H^+] = 0.04$$

$$\text{or } \frac{25 - \frac{307.5}{M}}{500} = 0.04$$

$$\text{or } 25 - \frac{307.5}{M} = 20$$

$$\text{or } \frac{307.5}{M} = 5 \Rightarrow M = \frac{307.5}{5} = 61.5 \text{ g.}$$

$$\therefore \text{Atomic mass of B} = 61.5 - 16 - 1 = \boxed{44.5}$$

$$\therefore \text{Molar mass of BCl} = 61.5 - 17 + 35.5 = 80 \text{ g.}$$

$$a = 2 (r_{B^+} + r_{A^-}) = 2 (0.8 + 1.2) = 4 \text{ \AA}$$

$$d = \frac{Z \times M}{N_A \times a^3} = \frac{4 \times 80}{6 \times 10^{23} \times (4 \times 10^{-8})^3} = \boxed{8.33 \text{ g/ml}}.$$

3. Ans ( 2450.53 or 2450.54 )

$$\Delta H^{\circ}_{27^{\circ}\text{C}} = -100 + 2(-60) - (-20) = -200 \text{ K cal mol}^{-1}$$

Reactants at

1 bar  $27^{\circ}\text{C}$

$\text{CH}_4(\text{g}) = 1 \text{ mole}$

$\text{O}_2 = 2.4 \text{ mole}$

$\text{N}_2 = 9.6 \text{ mole}$

Products at 1 bar TK

$\text{CO}_2(\text{g}) = 1 \text{ mole}$

$\text{H}_2\text{O}(\text{g}) = 2 \text{ mole}$

$\text{O}_2(\text{g}) = 0.4 \text{ mole}$

$\text{N}_2(\text{g}) = 9.6 \text{ mole}$

$$C_{P_m \text{ O}_2} = C_{P_m \text{ N}_2} = C_{P_m \text{ CO}_2} = \frac{7}{2}R$$

$$C_{P_M \text{ H}_2\text{O}(\text{g})} = 4R$$

$$200 \times 1000 = 1 \times \frac{7}{2}R \times (T - 300) + 2 \times 4R(T - 300) + 10 \times \frac{7}{2}R(T - 300)$$

$$\text{or } 200 \times 1000 = (T - 300) [3.5R + 8R + 35R]$$

$$\text{or } 200 \times 1000 = (T - 300) \times 2[46.5]$$

$$\text{or } T - 300 = \frac{100 \times 1000}{46.5} = \frac{20 \times 1000}{9.3} = 2150.53$$

$$\text{or } T = 2450.53 \text{ K.}$$

4. Ans ( 159.92 )

$$\Delta C_P = (C_{P_m \text{ CO}_2} + 2C_{P_m \text{ H}_2\text{O}}) - (C_{P_m \text{ CH}_4} + 2C_{P_m \text{ O}_2})$$

or

$$\Delta C_P = \left( \frac{7}{2}R + 2 \times 4R \right) - \left( 4R + 2 \times \frac{7}{2}R \right)$$

$$= 11.5 R - 11 R = 0.5 R$$

$$\Delta H_{T_2} = \Delta H_{T_1} + \Delta C_P (T_2 - T_1)$$

$$= -200 + 0.5R \left( \frac{400 - 300}{1000} \right)$$

$$= -200 + 0.1 = -199.90 \text{ Kcal}$$

$$\therefore q_H = 199.9 \text{ Kcal}$$

$$\eta_{\max} = \frac{T_H - T_L}{T_H} = \frac{400 - 300}{400} = 0.25$$

$$\eta = 0.8 \quad \eta_{\max} = 0.8 \times 0.25 = 0.20$$

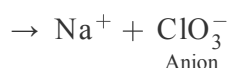
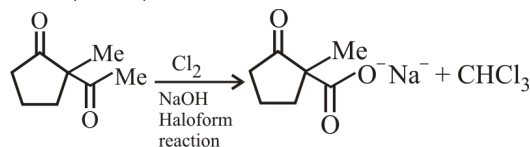
$$\text{or } \frac{|W|}{q_H} = 0.2$$

$$\text{or } |W| = 199.9 \times 0.2 = 39.98$$

$$\therefore q_H = |q_L| + |W|$$

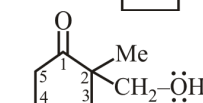
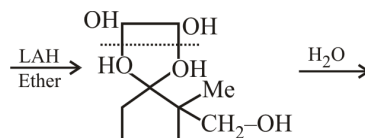
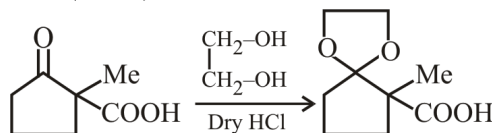
$$\text{or } |q_L| = 199.9 - 39.98 = 159.92 \text{ Kcal.}$$

5. Ans ( 3.00 )



R.S. of anion is 3

6. Ans ( 5.00 )



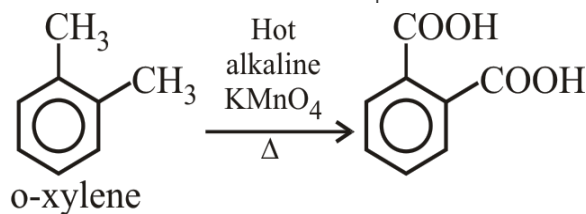
(According to IUPAC rule)

## PART-2 : CHEMISTRY

### SECTION-II (ii)

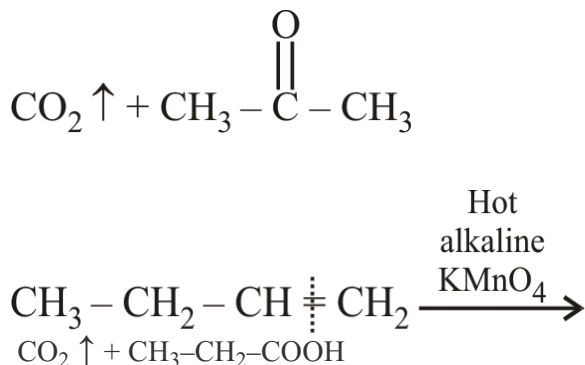
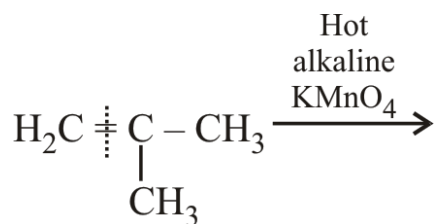
8. Ans ( 3 )

Ortho, meta and para xylene gives dicarboxylic acid on treatment with  $\text{KMnO}_4$



9. Ans (2)

Similar for M-xylene and P-xylene



### PART-3 : MATHEMATICS

#### SECTION-I (i)

1. Ans (A,B,C)

For nilpotent matrix of order 2,  $A^2 = 0$

$$A(I + A) = A + A^2 = A + 0 = A$$

for  $n \geq 2$

$$A(I + A)^n = A[I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n]$$

$$\Rightarrow A(I + A) = A + A^2[{}^nC_1 I + {}^nC_2 A + \dots] = A$$

similarly for  $A(I - A)^n$

For idempotent matrix,

$$A^2 = A = A^3 = \dots = A^n$$

$$(I + A)^n = I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n$$

$$= I + A(2^n - 1)$$

2. Ans (A,B)

$$z = \ln \tan \frac{t}{2}$$

$$\frac{dz}{dt} = \operatorname{cosec} t$$

$$\frac{dy}{dz} = \sin t \frac{dy}{dt}$$

$$\frac{d^2 y}{dz^2} = \frac{d}{dt} \left( \sin t \frac{dy}{dt} \right) \sin t$$

$$\Rightarrow \frac{d^2 y}{dt^2} + \cot t \frac{dy}{dt} + 4y \operatorname{cosec}^2 t = 0$$

3. Ans (A,D)

$$f(x) = \int_x^{x+\frac{\pi}{3}} |\sin \theta| d\theta \quad x \in [0, \pi]$$

$$f'(x) = \left| \sin \left( x + \frac{\pi}{3} \right) \right| - |\sin x| = 0$$

$$\Rightarrow \left| \sin \left( x + \frac{\pi}{3} \right) \right| = |\sin x|$$

$$x = \frac{n\pi}{2} - \frac{\pi}{6} \quad \left\{ x = \frac{\pi}{3}, \frac{5\pi}{6} \right\}$$

$$f(0) = \int_0^{\pi/3} \sin \theta d\theta = \frac{1}{2},$$

$$f(\pi) = - \int_{\pi}^{4\pi/3} \sin \theta d\theta = \frac{1}{2}$$

$$f\left(\frac{\pi}{3}\right) = \int_{\pi/3}^{2\pi/3} \sin \theta d\theta = 1,$$

$$f\left(\frac{5\pi}{6}\right) = \int_{5\pi/6}^{\pi} \sin \theta d\theta + \int_{\pi}^{7\pi/6} -\sin \theta d\theta = 2 - \sqrt{3}$$

4. **Ans (B,C,D)**

$$3f(x+2y) = f(2x) + f(y) + 2x + 10y + 1 \quad \forall x, y \in \mathbb{R}$$

$$\text{Let } x = y = 0$$

$$\Rightarrow f(0) = 1 \Rightarrow f'(0) = 2$$

Differentiate (1) w.r.t. y

$$6f'(x+2y) = f'(y) + 10$$

$$\text{Now let } y = 0$$

$$\Rightarrow 6f'(x) = f'(0) + 10$$

$$\Rightarrow f'(x) = 2$$

$$\Rightarrow f(x) = 2x + 1 \quad (\because f(0) = 1)$$

5. **Ans (A,B,C)**

$$(\sin^{-1}x)^2 - (\cos^{-1}x)^2 = \frac{\pi}{2} (\sin^{-1}x - \cos^{-1}x)$$

$$= \frac{\pi}{2} \left( \frac{\pi}{2} - 2\cos^{-1}x \right) \in \left[ -\frac{3\pi^2}{4}, \frac{\pi^2}{4} \right] \subset (-8, 3)$$

$$\text{Also, } \sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$$

$$\Rightarrow \sin^{-1}x + 2(\pi - \cos^{-1}x) = 2\pi$$

$$\Rightarrow \sin^{-1}x - 2 \left( \frac{\pi}{2} - \sin^{-1}x \right) = 0$$

$$\Rightarrow \sin^{-1}x = \frac{\pi}{3} \Rightarrow x = \frac{\sqrt{3}}{2} > \frac{1}{2}$$

6. **Ans (A,C)**

$$f_1(x) = e^x - \sqrt{-x} \Rightarrow f_1'(x) = e^x + \frac{1}{2\sqrt{-x}} > 0$$

$$f_2(x) = \sqrt{-x} + |x| = \sqrt{-x} - x \quad (\because x < 0)$$

$$\Rightarrow f_2'(x) = \frac{-1}{2\sqrt{-x}} - 1 < 0$$

$$f_3(x) = e^{1-x} - \sqrt{x-1}$$

$$\Rightarrow f_3'(x) = -e^{1-x} - \frac{1}{2\sqrt{x-1}} < 0$$

**PART-3 : MATHEMATICS**

**SECTION-I (ii)**

7. **Ans (B)**

$$f'(x) = \sqrt{2f(x) - f^2(x)}$$

$$\int \frac{dy}{\sqrt{1 - (y-1)^2}} = \int dx$$

$$\sin^{-1}(y-1) = x + c$$

$$x = 0, y = 1, c = 0$$

$$\sin^{-1}(y-1) = x$$

$$\boxed{y = 1 + \sin x}$$

$$\int_0^{\pi} (1 + \sin x) dx = \pi + 2$$

8. **Ans (A)**

$$\int_0^{\pi/2} \frac{\sin x dx}{1 + \cos^2 x}$$

$$\cos x = t$$

$$= - \int_1^0 \frac{dt}{1+t^2} = -(\tan^{-1}t)_1^0 = +\frac{\pi}{4}$$

9. **Ans (C)**

$$x^{\ell nx} = e^{2x}$$

$$\ell n(x^{\ell nx}) = \ell n(e^{2x})$$

$$\ell nx \ell nx = 2 + \ell nx$$

$$(\ell nx - 2)(\ell nx + 1) = 0$$

$$x = e^2, \quad x = \frac{1}{e}$$

$$\beta = e^2 \quad \alpha = \frac{1}{e}$$

10. **Ans (B)**

$$\lim_{x \rightarrow e^2} \frac{x^{\ell nx} - ce^2}{e^2x - e^4} \quad \left( \frac{0}{0} \text{ form} \right)$$

$$\Rightarrow \boxed{c = e^2}$$

$$\lim_{x \rightarrow e^2} \frac{2x^{\ell nx-1} \ell nx}{e^2} = 4$$

## PART-3 : MATHEMATICS

### SECTION-II (i)

1. Ans ( 3.00 )

$$A^n - A^{n-2} = A^2 - I$$

$$A^{50} - A^{48} = A^2 - I$$

$$A^{48} - A^{46} = A^2 - I$$

$\vdots$

$$A^4 - A^2 = A^2 - I$$

---


$$A^{50} - A^2 = 24A^2 - 24I$$


---

$$A^{50} = 25A^2 - 24I = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$

$$|A^{50}| = 1$$

$$A^{50}U_1 = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\Rightarrow U_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \text{ Similarly } U_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$U_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now verify

2. Ans ( 53.00 )

$$A^n - A^{n-2} = A^2 - I$$

$$A^{50} - A^{48} = A^2 - I$$

$$A^{48} - A^{46} = A^2 - I$$

$\vdots$

$$A^4 - A^2 = A^2 - I$$

---


$$A^{50} - A^2 = 24A^2 - 24I$$


---

$$A^{50} = 25A^2 - 24I = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$

$$|A^{50}| = 1$$

$$A^{50}U_1 = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 25 \\ 25 \end{bmatrix}$$

$$\Rightarrow U_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \text{ Similarly } U_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$U_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now verify

3. Ans ( 0.00 )

$$f(x) = 4^x, g(x) = \sin(\cos(\cos \pi x) + \cos(\sin \pi x))$$

$$\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)} = 0, g(-x) = g(x)$$

$$g(x) \text{ is periodic with period } \frac{1}{2}, g'(0) = 0$$

4. Ans ( 255.00 )

$$\int_0^4 f(x) \cdot g(x) dx = \lambda \int_0^{1/2} f(x) \cdot g(x) dx$$

$$\frac{\int_0^{1/2} f \cdot g dx + \int_{1/2}^1 f \cdot g dx + \int_1^{3/2} f \cdot g dx + \dots + \int_{7/2}^4 f \cdot g dx}{\int_0^{1/2} f \cdot g dx} = \lambda$$

in second integration

$$\int_{1/2}^1 f \cdot g dx \quad \text{put } x = t + \frac{1}{2}$$

$$= \int_0^{1/2} 4^{t+1/2} g(t) dt = 2 \int_0^{1/2} f \cdot g dx$$

Similarly in third  $x = t + 1$

$$\Rightarrow \lambda = 1 + 2 + 2^2 + 2^3 + \dots + 2^7$$

$$\Rightarrow 2^8 - 1 = 255$$

5. Ans ( 51.00 )

$$\text{Let } f(x) = (x-1)^2(ax^2 + bx + c)$$

$$\Rightarrow a + b + c = 2 \text{ \& } f'(c) = -8, f'(2) = 8$$

$$\Rightarrow c = 3, a = 1, b = -2$$

$$\Rightarrow f(x) = (x-1)^2(x^2 - 2x + 3)$$

$$g(x) = 3(x-1)^2 \Rightarrow g'(0) = -6, f'(0) = -8$$

Angle of intersection

$$= \tan^{-1} \left( \frac{2}{1+48} \right) = \tan^{-1} \frac{2}{49}$$

6. Ans ( 4037.00 )

$$\text{Length of subtangent} = \left| \frac{y}{dy/dx} \right| = \left| \frac{y}{6(x-1)} \right|$$

$$\left| \frac{3(x-1)^2}{6(x-1)} \right| = 2018 \Rightarrow \frac{x-1}{2} = \pm 2018$$

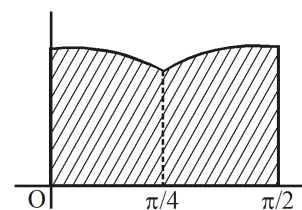
$$x = 4037 \text{ or } x = -4035$$

## PART-3 : MATHEMATICS

### SECTION-II (ii)

7. Ans ( 200 )

Required Area



$$= \int_0^{\pi/4} \cos x dx + \int_{\pi/4}^{\pi/2} \sin x dx$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

8. Ans ( 6 )

$$\because (x, 1), (y, 2), (z, 1), (2, 3) \in A \times B$$

$$\Rightarrow A = \{x, y, z, 2\}$$

$$B = \{1, 2, 3\}$$

$$\text{Number of onto map} = {}^4C_2 \cdot 3! = 36$$

9. Ans ( 4 )

$$Q^T P = P^T$$

$$P^T Q = P$$

$$PP^T Q = P^2$$

$$Q = P^2$$

$$\Rightarrow Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \Rightarrow Q^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 & -\lambda \\ \lambda & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\sqrt{2} \\ \sqrt{2} & 0 \end{bmatrix}$$

$$\therefore \lambda = \sqrt{2}$$

$$\text{Now } \because \sqrt{2}(I + Q^T) = 2P^T$$

$$\Rightarrow \text{adj.}(I + Q^T) \cdot \text{adj.}(\lambda P)$$

$$= \text{adj.}(\sqrt{2}P^T) \cdot \text{adj.}(\sqrt{2}P)$$

$$= 2\text{adj.}(P^T P) = 2I$$